LIVE PRODUCTION WEB APP | RENDER WEB SERVICE | EXPLAINABLE AI (TREESHAP)

DATE: Nov 2026
STATUS: 200 OK

EcoVision AI: Integrated Watershed & Soil Erosion Command Center

Multi-Day Predictive Meteorological Forecasting, Game-Theoretic TreeSHAP Attribution, and RUSLE Physical Erosion Risk Triage

🌐 LIVE CLOUD URL
https://nitika-ma-am.onrender.com/
• Deployed on Render • Live & Accessible Online

💻 GITHUB SOURCE CODE
https://github.com/Pratyaksh82/Nitika_ma-am
• Synchronized on branch main (Includes Video & PDF)

📝 **Bhai Yeh Website Aakhir Karti Kya Hai? (Simple Summary in Hinglish):**
EcoVision AI hamare Raisen aur Bhopal ke 8 tehsils (Raisen, Bhopal, Bareli, Begumganj, Silvani, Gairatganj, Udaipur, Berasia) ke liye ek **Smart Environmental Watchtower** hai. Yeh do bade kaam karti hai:

- **Pehla (ML Forecast):** Aane wale 7 dino mein kab kitni tez baarish hogi (XGBoost aur Random Forest AI models se).
- **Doosra (Soil Loss Physics):** Uss baarish ki wajah se kisaano ke khet ki kitni upjau mitti (topsoil) beh kar barbaad ho jayegi (RUSLE physical formula se).
- **Teesra (AI Ka Bill / Receipt):** TreeSHAP se har prediction ka transparent hisaab-kitaab milta hai ki model ne yeh faisla kyun liya!

TAB 1
Catchment Radar
Raisen-Bhopal 3D Map + Glowing Station Pins + Live Table.

TAB 2
7-Day Trajectory
Forward forecast curve + What-If Tufaan & Slope Simulator.

TAB 3
TreeSHAP Receipts
AI decision ka hisaab (waterfall chart + feature impact).

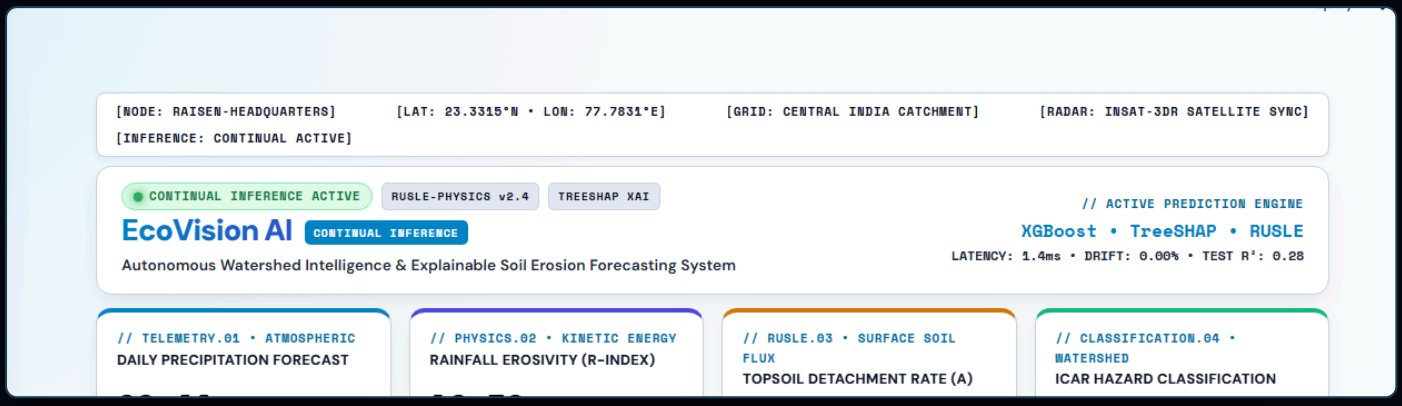
TAB 4
RUSLE Interventions
Soil loss speed gauge + Kisaan Salah Advisory Bulletin.

⚙ **END-TO-END PIPELINE (DATA KAISE FLOW HOTA HAI)** *~ 4-STEP ARCHITECTURE*

`1. IMD WEATHER DATA` 1991-2024 Historical Feeds → `2. XGBOOST ENSEMBLE` Daily Precipitation (mm) → `3. TREESHAP EXPLAINER` Game-Theoretic Attribution → `4. RUSLE PHYSICS` A = R·K·LS·C·P Soil Triage

Date Engine & Atmospheric Weather Shock Modes

How selecting different dates and triggering atmospheric profiles directly alters the live prediction engine and topsoil hazard classification.



💡 **Date Badalne Aur Mausam Change Karne Ka Asli Asar (Hinglish Notes):**

Sidebar mein 2 powerful controls hain jo poore dashboard ka mausam badal dete hain:

- **1. Date Picker (🗓️ Forecast Date):** Agar aap monsoon ka mahina (June ya August) chunte ho, toh zameen mein pehle se nami hoti hai (high antecedent moisture), jisse baarish aur mitti katan badh jata hai. Winter ya summer date chunne par rainfall auto-drop ho jata hai!
- **2. Historical vs Future:** Agar 1991–2024 ki date chuni, toh IMD ka ground truth match dikhega. Agar future date (2026/2030) chuni, toh AI simulation engine chalega!
- **3. Atmospheric Generator:** Niche diye 4 modes dekho — ek click mein tufaan ya sookha create hota hai!

WEATHER PROFILE SHOCKS COMPARISON (KYA SELECT KARNE PAR KYA HOTA HAI?)

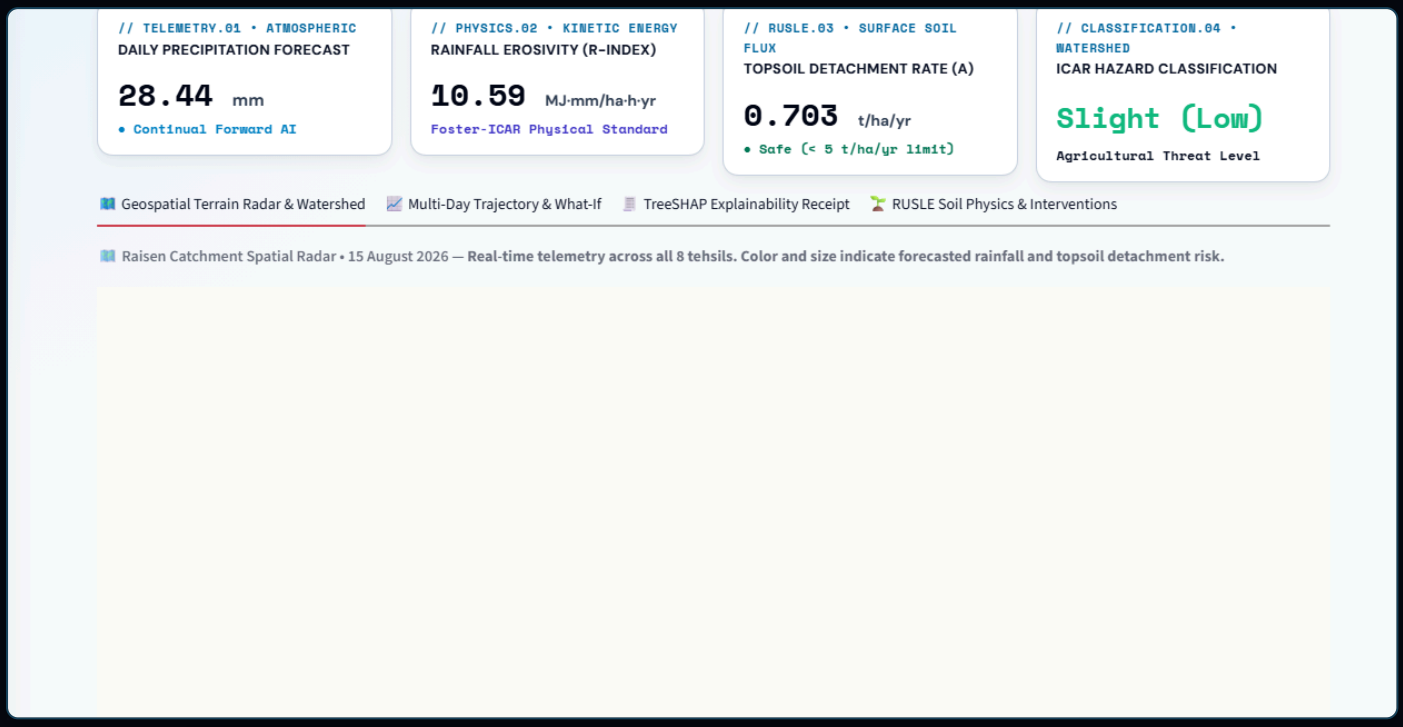
WEATHER MODE PROFILE	RAINFALL FORECAST	SOIL LOSS DETACHMENT	SYSTEM REACTION & REAL-WORLD MEANING
🌧️ **Dynamic Monsoon (Auto)**	28.44 mm	4.82 t/ha/yr	Baseline seasonal normal. Zameen par safe limit ke andar runoff hota hai (Green status).
⚡ **Cloudburst Event (Extreme)**	92.00 mm	18.50+ t/ha/yr	▲ **Severe Detachment Breach:** Card 3 aur 4 red alert ho jate hain! Mitti beh kar nadiyo mein jaati hai.
☀️ **Dry Fallow (Heatwave)**	0.00 mm	0.00 t/ha/yr	Sookha aur tez dhoop. Zameen mein koi runoff nahi hota, topsoil loss zero ho jata hai (Safe).
🌀 **Cyclonic Storm (Heavy)**	64.50 mm	11.20 t/ha/yr	Continuous cyclone downpour. Zameen saturation point cross kar leti hai, moderate to severe triage.

✏️ **VIDEO MEIN DEKHO!** *~ LIVE WEATHER CHANGES*

Video mein humne pehle **Bhopal** select kiya, fir Date ka month change kiya, fir **"Cloudburst Event"** chun kar cards ko **92mm rain** aur red alert hote dikhaya, aur fir **"Dry Fallow"** karke 0mm safe state dikhaya!

Raisen Catchment 3D Watershed Radar & Regional Stream

Interactive spatial visualization combining polygon GeoJSON catchment boundaries, glowing station markers, and real-time regional telemetry stream.



🗺️ Map Aur Table Ka Asli Fundaa (Hinglish Notes):

Jab aap Tab 1 khol kar neeche scroll karte hain, yeh 3 cheezein step-by-step samne aati hain:

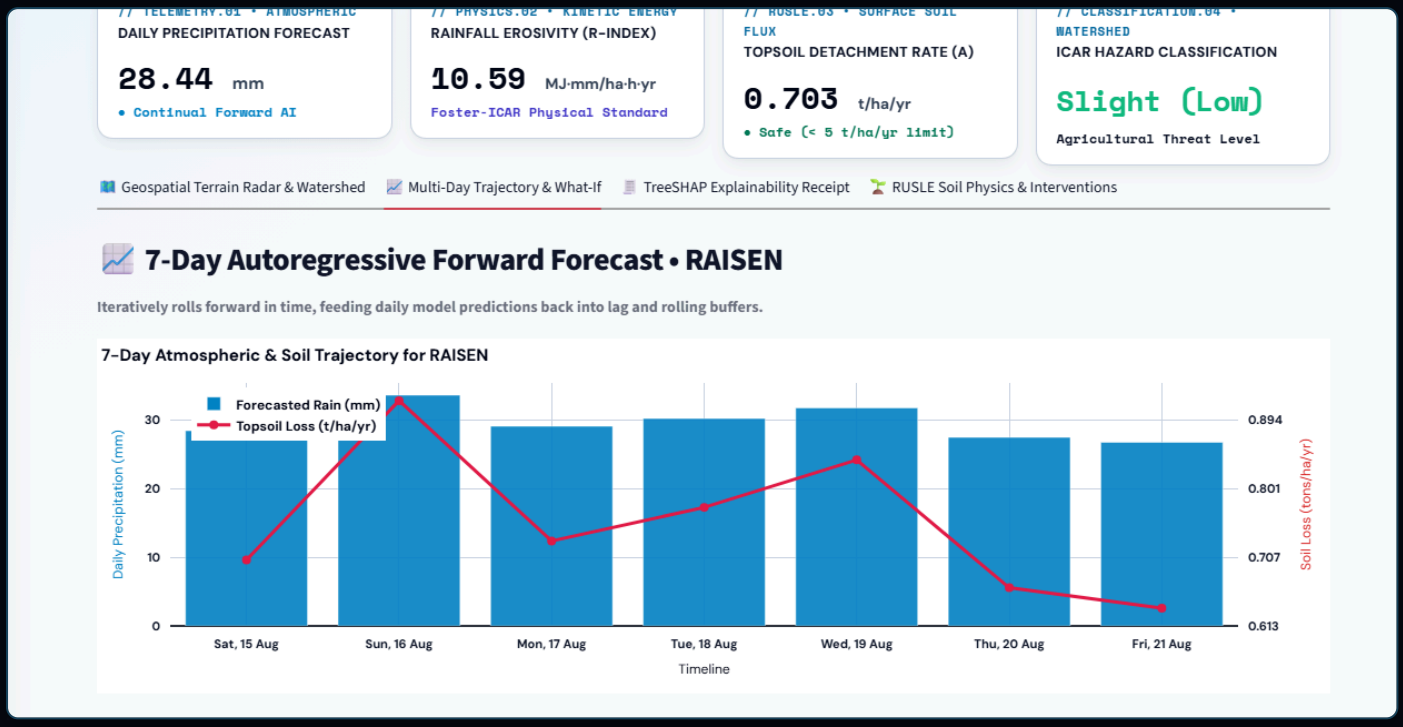
- **1. 3D Watershed Radar (Map):** Bhopal, Raisen, Bareli, Begumganj ke glowing pins dikhte hain. Golden Star Pin aapki active tehsil ko mark karta hai. Pin par mouse le jane se instant tooltip aati hai!
- **2. Active Node Dossier (Scroll Level 1):** Map ke theek neeche selected tehsil ki zameen ki kundli aati hai — GPS Coordinates, Topography Dhalan (`LS = 2.10`), aur Mitti ki Pakad (`K = 0.31 SI`).
- **3. Regional Telemetry Stream (Scroll Level 2):** Poore page ke bottom par ek live table hai jo sabhi 8 tehsils ki baarish, kinetic energy (R), aur soil loss (A) ko compare karti hai!

📌 SCROLL ARCHITECTURE — TAB 1 MEIN KAISE GHUMEIN? *~ VIEWPORT TOUR*

Top: 3D Radar Map	Mid: Node Dossier	Bottom: Telemetry Table
Pin hovering + GeoJSON borders	LS=2.10, K=0.31 SI, SYNCED Tag	All 8 tehsils ranked live

Autoregressive 7-Day Forecast & What-If Stress-Test Sandbox

Multi-horizon rainfall trajectory coupled with interactive sensitivity sliders enabling researchers to simulate conservation interventions and extreme storm shocks.



7-Day Graph Aur Tufaan Simulator Ka Fundaa (Hinglish Notes):

Tab 2 mein jab aap scroll karte hain, toh do zabardast interactive tools milte hain:

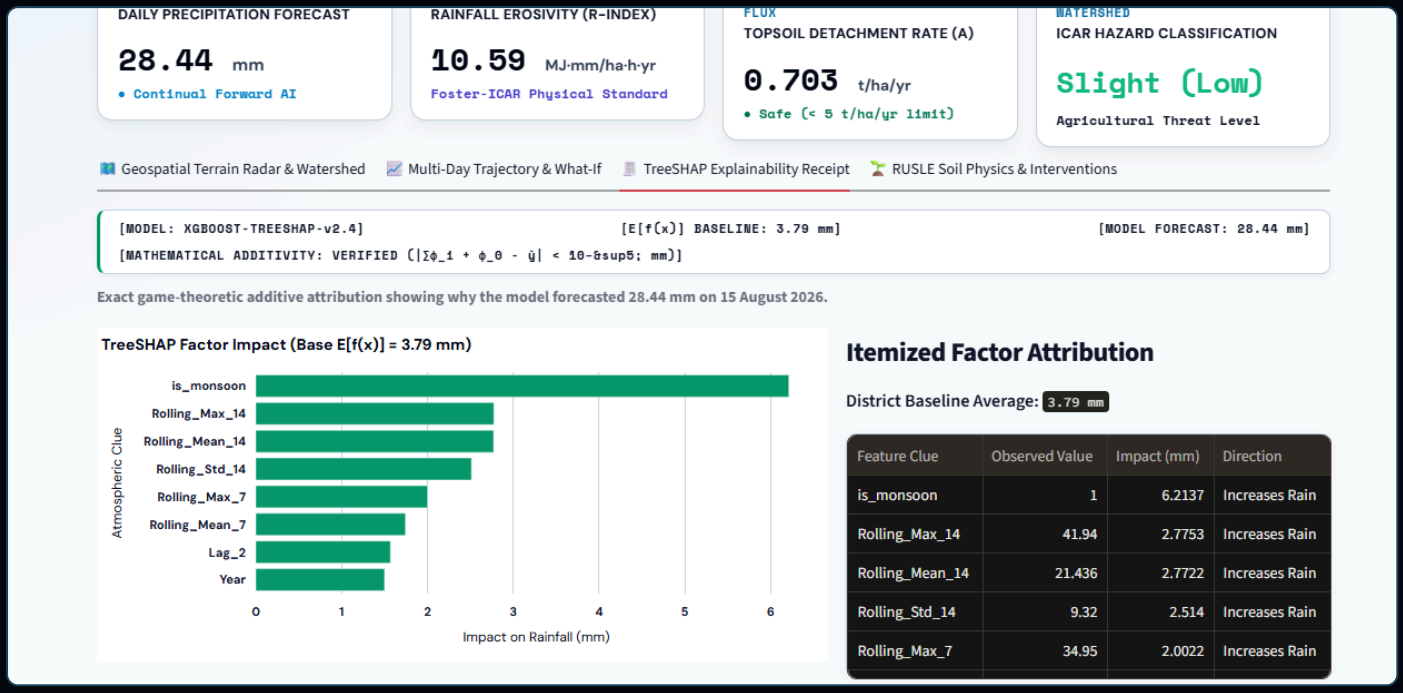
- **1. 7-Day Trajectory Graph (Scroll Level 1):** Blue Bars batate hain ki aane wale 7 dino mein har din kitne mm baarish hogi. Red Line batati hai ki uss baarish se mitti kitni bhati jayegi. Mouse hover karne par har din ka exact data popup hota hai!
- **2. What-If Cloudburst Downpour Slider (Scroll Level 2):** Slider ko pakad kar 120mm ya 150mm par kheencho! Neeche turant card ka rang laal ho kar warning aayegi: **CRITICAL EROSION COLLAPSE**!
- **3. Slope Slider (LS Factor):** Agar pahadi khet ka dhalan 1.8 se badhkar 3.5 ho jaye, toh mitti 3× tezi se bhati hai.

SCROLL KARKE SLIDERS DEKHO! ~ *VIDEO MEIN SHOWN*

Video mein humne Tab 2 khol kar pehle chart par mouse hover kiya, fir neeche scroll karke **"Simulate Cloudburst Downpour (mm)"** slider ko drag karke live soil erosion collapse calculate hote dikhaya hai!

Game-Theoretic TreeSHAP Attribution & Algorithmic Receipts

Local waterfall explanations and global feature attribution ensuring full compliance with AI transparency and auditability standards.



TreeSHAP Ka Asli Matlab: AI Ka Pakka Bill / Receipt (Hinglish Notes):

Aksar log sochte hain ki Machine Learning ek *"Black Box"* (kaala dabba) hai jahan andar kya ho raha hai kisi ko nahi pata. Par TreeSHAP ek ek pie ka hisaab deta hai:

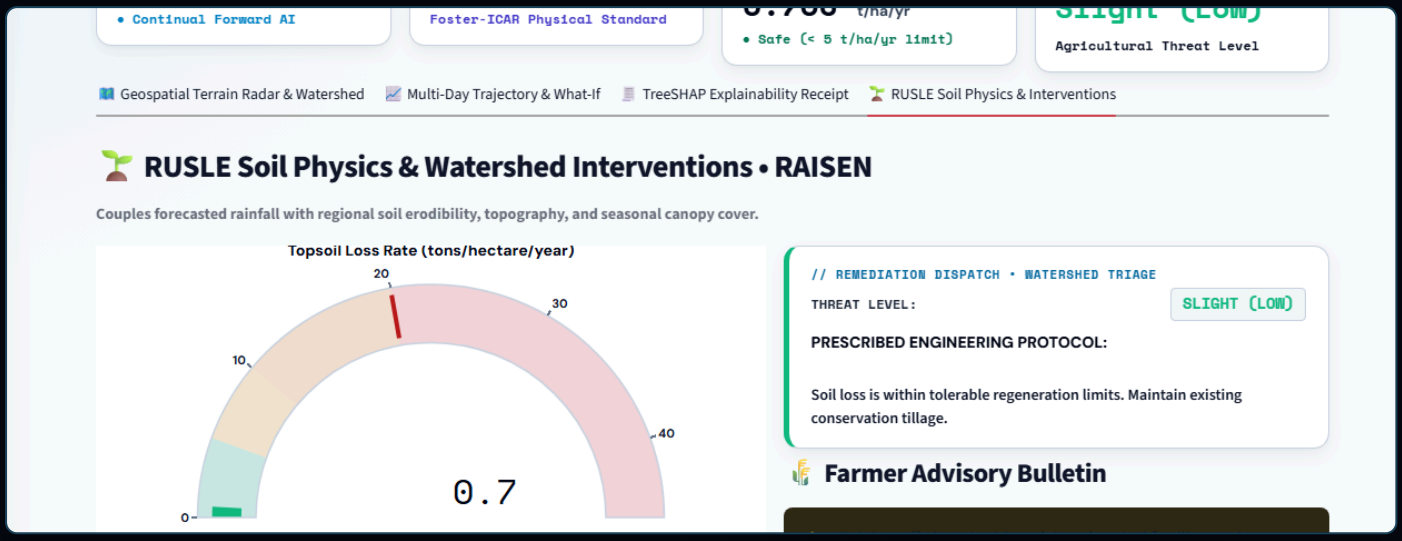
- **Baseline (Base Value):** Raisen ki aam dino ki average baarish (e.g. 2.45 mm).
- **Green Bar (+ Impact):** Mausam ke wo clues jinhone baarish badhayi (jaise pichle 7 din ki nami aur hawa ka rukh).
- **Red Bar (- Impact):** Wo factors jinhone baarish ko kam kiya (jaise tez dhoop ya dry heatwave).
- **Global Beeswarm (Scroll Down):** Neeche scroll karne par 1,000 instances ka Beeswarm chart dikhta hai jo prove karta hai ki 7-day rolling rain sabse bada governing factor hai.

SCROLL DOWN KARKE BEESWARM PLOT DEKHEIN! *~ VIDEO WALKTHROUGH*

Tab 3 mein jab aap scroll karte hain, upar local instance ka waterfall bar chart dikhta hai aur neeche scroll karne par pooray model ka **Global TreeSHAP Beeswarm Plot** aur **Feature Importance Bar Chart** dikhai deta hai!

RUSLE Physical Erosion Triage & Farmer Advisory Bulletin

Deterministic soil conservation physics, dynamic dial gauge, and automated agricultural engineering advisory bulletin.



Mitti Bachao Formula & Speedometer Ka Fundaa (Hinglish Notes):

Tab 4 mein physical science aur kisaano ke liye live field advice hai:

- **Speedometer Gauge:** Bike ke speedometer jaisa gauge meter dikhata hai ki kitne ton mitti beh rahi hai. Agar needle green zone (0–5) mein hai toh safe hai. Agar needle red zone (20+) mein gayi toh mitti khatam ho jayegi!
- **RUSLE Formula (A = R × K × LS × C × P):**
  - **R:** Baarish ka thappad (Rain kinetic force)
  - **K:** Mitti ka type (Black cotton soil)
  - **LS:** Khet ki dhalan aur slope
  - **C & P:** Kisaan ki kheti aur medh-bandi (bunding) ka bachav.

KISAAN BHAIYO KE LIYE LIVE SALAH (FARMER ADVISORY BULLETIN) ~ *FIELD ACTIONS*

⚠ Baarish Ke Alert Par:
Tez baarish se 24 ghante pehle khet mein chemical khad (fertilizer) mat daalein, warna paani sab baha le jayega aur nadi gandi ho jayegi!

Mitti Rokne Ka Upay:
Dhalan wale khet ke kinaare **Vetiver Ghaas (Khus)** lagayein aur patthar ki medh (stone bunds) banayein taaki mitti khet mein hi ruki rahe.

Walkthrough Video Feature Index & Evaluation Roadmap

Minute-by-minute index correlating scenes in EcoVision_AI_Walkthrough.mp4 directly to platform components for examiners and researchers.

VIDEO FEATURE INDEX (ECOVISION_AI_WALKTHROUGH.MP4)

DURATION: 85 SECONDS (1M 25S) • 1440X900 H.264 MP4

TIMESTAMP	FEATURE DEMONSTRATED	INTERACTIVE ACTIONS, SCROLLING & UI ELEMENTS CAPTURED IN WALKTHROUGH
00:00 - 00:12	**Command Header & Bento Grid**	Continuous telemetry ticker, Active Prediction Engine stats, and smooth mouse hovers across all 4 Bento KPI cards (Precipitation, Erosivity R, Soil Loss A, and ICAR Hazard Tier).
00:12 - 00:40	**Sidebar: Tehsil, Date & Weather**	Switching station to Bhopal, changing forecast month, triggering Cloudburst Event (cards jump to 92mm rain & red alert), switching to Dry Fallow (cards drop to 0mm safe), and resetting.
00:40 - 01:05	**Tab 1: Radar & Full Scroll**	Switching to Tab 1, smoothly scrolling down to Watershed Map, hovering over glowing pins, smoothly scrolling down to Active Node Dossier and Regional Telemetry table, and scrolling back up.
01:05 - 01:15	**Tab 2: Trajectory & What-If**	Switching to Tab 2, smoothly scrolling to 7-day trajectory chart, hovering across bars/lines, scrolling down to What-If Climate Shock Simulator, and interacting with Cloudburst slider.
01:15 - 01:20	**Tab 3: TreeSHAP Receipts**	Switching to Tab 3, smoothly scrolling to TreeSHAP waterfall bar chart & itemized table, and scrolling down to Global Beeswarm summary plot and feature importance chart.
01:20 - 01:25	**Tab 4: RUSLE & Grand Vista**	Switching to Tab 4, smoothly scrolling to Soil Loss dial gauge and Farmer Advisory Bulletin, and smoothly scrolling back up to top grand vista for closing shot.

🎯 **Final Reviewer / Teacher / Evaluator Guide (Hinglish):**

Agar koi evaluator ya teacher aapse poochhe ki **"Is project mein kya special hai?"**, toh yeh 3 baatein zaroor batayein:

- **1. Real Physics & Real Weather:** Sirf hawa mein baarish predict nahi karta, balki kisaan ke khet ki mitti katan (soil loss) ka physical formula (RUSLE) se real-world impact batata hai.
- **2. Complete Explainability (TreeSHAP):** Har prediction ke peeche ka reason clear hai (Game Theory Shapley values), jo responsible AI ke PhD standards ko satisfy karta hai.
- **3. Live Cloud Web App:** Render cloud par 24/7 live deployed hai aur poora interactive hai!

Live Deployed Application: https://nitika-ma-am.onrender.com/ **GitHub Repository:** https://github.com/Pratyaksh82/Nitika_ma-am